1.7 Lebesgue Measure

With the notion of Lebesgue outer measure, we can now establish the notion of Lebesgue measure.

A set E is said to be **measurable** if, for any set A we have $m^*(A) = m^*(A \cap E) + m^*(A \cap (R - E))$. In other words, the outer measure of A equals to the sum of the outer measure of the part inside E and the part outside E.

If a set E is measurable, we simply use its outer measure as its **Lebesgue measure**. That is, $m(E) = m^*(E)$.

We will show that the Lebesgue measure possesses all the desired properties

- Interval Measure
- Monotonicity
- Translation Invariant
- Countable Additivity

We will show that the set of all measurable sets form a $\sigma-$algebra M. And M contains the Borel $\sigma-$algebra B . Moreover, we will show that $B \subset M$, i.e. there are measurable sets that are not Borel. Finally, there are also sets that are not measurable.

Properties of Lebesgue Measure

Proposition 1.7.1: The empty set $\emptyset$ is measurable. Thus, $m(\emptyset) = 0$. *Proof:* For any set A, we have $m^*(A \cap \emptyset) = m^*(\emptyset) = 0$. Therefore, $m^*(A) = m^*(A \cap (R - \emptyset)) = m^*(A \cap (R - \emptyset)) + m^*(A \cap \emptyset)$. **QED**

Proposition 1.7.2: Any countable set is measurable, and its measure is 0. *Proof:* Let E be a countable set. For any set A, $A \cap E$ is a subset of E and thus also countable. So, $m^*(A \cap E) = 0$. Therefore,

$$m^*(A) \geq m^*(A \cap (R - E)) = m^*(A \cap (R - E)) + m^*(A \cap E)$$

. **QED**

Proposition 1.7.3: If E, F are both measurable and $E \subseteq F$, then $m(E) \leq m(F)$. *Proof:* This is obvious. **QED**

In what follows, we show that the set of all measurable sets form a $\sigma-$algebra.

Proposition 1.7.4: If E is measurable, then $R - E$ is also measurable. *Proof:* For any set A, since E is measurable we have

$$(1) \quad m^*(A) = m^*(A \cap E) + m^*(A \cap (R - E))$$

. Note that $E = R - (R - E)$. Therefore, (1) exactly shows that $R - E$ is measurable. **QED**

To show that a countable union of measurable sets is measurable, we need the following lemmas.

Lemma 1.7.5: Any finite union of measurable sets is measurable. *Proof:* It suffices to prove the case of 2. Let E_1, E_2 be two measurable sets. We would like to show $E_1 \cup E_2$ is also measurable. For any set A, since E_1 is measurable, we have

$$(1) \quad m^*(A) = m^*(A \cap E_1) + m^*(A \cap (R - E_1))$$

. Since E_2 is measurable, we can decompose $m^*(A \cap (R - E_1))$ into

$$(2) \quad m^*(A \cap (R - E_1)) = m^*(A \cap (R - E_1) \cap E_2) + m^*(A \cap (R - E_1) \cap (R - E_2))$$

. Note that the union of $A \cap E_1$ and $A \cap (R - E_1) \cap E_2$ is exactly $A \cap (E_1 \cup E_2)$. And $A \cap (R - E_1) \cap (R - E_2)$ is exactly $A \cap (R - (E_1 \cup E_2))$. By the sub-additivity of outer measure, we have

$$(3) \quad m^*(A \cap E_1) + m^*(A \cap (R - E_1) \cap E_2) \geq m^*(A \cap (E_1 \cup E_2))$$

. Combining (1)(2)(3), we obtain the conclusion as desired. **QED**

Lemma 1.7.6: Let $E_1, \ldots E_n$ be a finite disjoint collection of measurable sets. Then for any set A, we have $m^*(A \cap (\bigcup_{k=1}^n E_k)) = \sum_{k=1}^n m^*(A \cap E_k)$

Proof: For convenience, we let $m_k = m^*(A \cap E_k)$, and let $p_k = m^*((A \cap \bigcup_{j=1}^n E_j) - (\bigcup_{j=1}^k E_j))$. Since $E_1, \ldots, E_n$ are all measurable and disjoint, we have

$$\begin{aligned} m^*(A \cap (\bigcup_{k=1}^n E_k)) &= m_1 + p_1 \\ &= m_1 + m_2 + p_2 \\ &\ldots \\ &= m_1 + \ldots + m_n \end{aligned}$$

. Note that $m_1 + \ldots + m_n$ is $\sum_{k=1}^n m^*(A \cap E_k)$. **QED**

Lemma 1.7.7: For any countable collection of measurable sets $E_1, \ldots, E_n, \ldots$, there exists a countable collection of disjoint measurable sets $E'_1, \ldots, E'_n, \ldots$ such that $\bigcup_{n=1}^\infty E_n = \bigcup_{n=1}^\infty E'_n$. *Proof:* Take $E'_1 = E_1$, and $E'_n = E_n - \bigcup_{k=1}^{n-1} E_k$ for $n \geq 2$. Clearly, the two unions are equal. In addition, the measurability of E'_n follows from Proposition 1.7.4, Lemma 1.7.5 and the fact that $E_1, \ldots, E_n, \ldots$ are all measurable. **QED**

With these lemmas, we have **Proposition 1.7.8: The union of a countable collection of measurable sets is also measurable.** *Proof:* Let $E_1, \ldots, E_n, \ldots$ be a countable collection of measurable sets, and let $E = \bigcup_{n=1}^\infty E_n$. By Lemma 1.7.7, we may assume they are mutually disjoint. Let $F_n = \bigcup_{k=1}^n E_k$. By Lemma 1.7.5, F_n is measurable for any n. So, by Proposition 1.7.4, $R - F_n$ is also measurable. Note that $F_n \subseteq E$ for any n. So, pick a n and for any set A we have

$$(1) \quad m^*(A) = m^*(A \cap F_n) + m^*(A \cap (R - F_n))$$

. By Lemma 1.7.7, we have

$$(2) \quad m^*(A \cap F_n) = \sum_{k=1}^n m^*(A \cap E_k)$$

. Since $R - E \subseteq R - F_n$, we have

$$(3) \quad m^*(A \cap (R - F_n)) \geq m^*(A \cap (R - E))$$

. Combining (1)-(3), we have

$$(4) \quad m^*(A) \geq \sum_{k=1}^n m^*(A \cap E_k) + m^*(A \cap (R - E))$$

. (4) holds for arbitrary large n. Therefore, we must have

$$(5) \quad m^*(A) \geq \sum_{k=1}^{\infty} m^*(A \cap E_k) + m^*(A \cap (R - E))$$

. Finally, note that

$$(6) \quad \sum_{k=1}^{\infty} m^*(A \cap E_k) \geq m^*(A \cap E)$$

. Combining (5) and (6), we obtain the conclusion as desired. **QED**

With these propositions and the fact that R is also measurable, we can easily obtain **Proposition 1.7.9: The set of all measurable sets form a $\sigma-$algebra.** *Proof:* Simple. **QED**

In what follows we show the $\sigma-$algebra of measurable sets contains the Borel $\sigma-$algebra. In other words, every Borel set is measurable.

To this end, it suffices to show **Proposition 1.7.10: Every interval of the form (a, ∞) is measurable.** *Proof:* Consider any set A. We would like to show

$$(1) \quad m^*(A) \geq m^*(A \cap (a, \infty)) + m^*(A \cap (-\infty, a])$$

.To this end, not that $(-\infty, a)$ and (a, ∞) are open sets. So for any $x \in A \cap (a, \infty)$, there exists $O_x \subseteq (a, \infty)$ that covers x. Therefore, there exists an open cover $\{O_x\}_{x \in A \cap (a, \infty)}$ of $A \cap (a, \infty)$ such that $O_+ = \bigcup O_x \subseteq (a, \infty)$.

Similarly, there is an open set $O_- \subseteq (-\infty, a)$ that covers $A \cap (-\infty, a)$. Finally, we pick an open interval $I_n = (a - \frac{1}{n}, a + \frac{1}{n})$ that covers $A \cap \{a\}$. To justify (1), it suffices to show that for any open cover O of A, where $O \subseteq O_+ \cup O_- \cup I_n$, we have

$$(2) \quad m^*(O) \geq m^*(A \cap (a, \infty)) + m^*(A \cap (-\infty, a])$$

. Observe that every such O will be the union of three parts: $O_1 \subseteq O_+$, $O_2 \subseteq O_-$ and $I' \subseteq I_n$. In addition, O_1 covers $A \cap (a, \infty)$, O_2 covers $A \cap (-\infty, a)$, and I' covers a. **QED**

Corollary 1.7.11: Every Borel set is measurable. *Proof:* Simple. **QED**

Finally, we show the translation invariant and countable additivity of Lebesgue measure.

Proposition 1.7.12: If E is measurable, so is $E + \lambda$. *Proof:* Consider any set A. Recall that outer measure has translation invariant property. So $m^*(A \cap (E + \lambda)) = m^*((A - \lambda) \cap E)$. Similarly, $m^*(A \cap (R - (E + \lambda))) = m^*((A - \lambda) \cap (R - E))$. Since E is measurable, for the set $A - \lambda$ we have

$$m^*(A - \lambda) = m^*((A - \lambda) \cap E) + m^*((A - \lambda) \cap (R - E))$$

. Finally, note that $m^*(A) = m^*(A - \lambda)$, we obtain the conclusion as desired. **QED**

Proposition 1.7.13 (Countable Additivity): $E_1, \ldots, E_n, \ldots$ are countable, measurable and disjoint. Then,

$$m(\bigcup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} m(E_n)$$

Proof: Let $E = \bigcup_{n=1}^{\infty} E_n$.

By the sub-additivity of outer measure, we have

$$(1) \quad m(E) \leq \sum_{n=1}^{\infty} m(E_n)$$

. On the other hand, by Lemma 1.7.6, the finite additivity holds and we have

$$(2) \quad m(\bigcup_{n=1}^{k} E_n) = \sum_{n=1}^{k} m(E_n)$$

. Again, by the sub-additivity of outer measure, since $\bigcup_{n=1}^{k} E_n \subseteq E$ we have

$$(3) \quad m(E) \geq m(\bigcup_{n=1}^{k} E_n)$$

(3) holds for arbitrarily large k, and thus it must hold for ∞. By combining (1)-(3) we have our desired result.
QED